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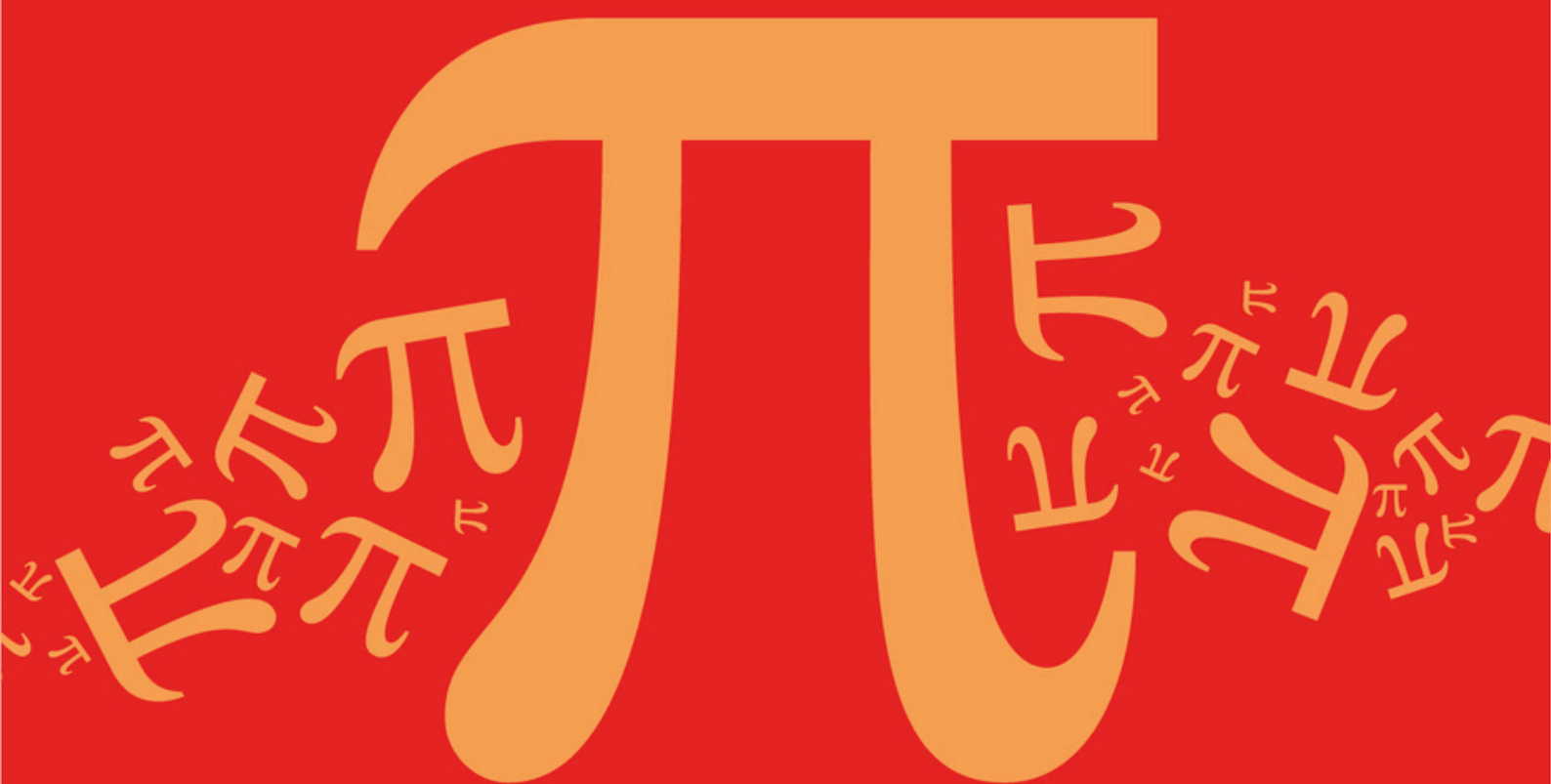
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AQA A Level Further Mathematics (7367)



Topic

C: Matrices

Sub topic

C3: Matrix Transformations (3D)

Topic Questions

AQA A Level Further Mathematics (7367)

Question Paper

Topic

C: Matrices

Sub topic

C3: Matrix Transformations (3D)

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**To be used by all students and teachers for
AQA A Level Further Mathematics (7367)
for exams from 2025 onwards.**

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Name : _____

Date : _____

Time : _____

Total Marks Available : _____

Total Marks Received : _____

Q1.

$$\begin{bmatrix} \frac{1}{2} & -\frac{\sqrt{3}}{2} & 0 \\ \frac{\sqrt{3}}{2} & \frac{1}{2} & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Describe fully the transformation given by the matrix

(Total 3 marks)

Q2.

The linear transformations T_1 and T_2 are represented by the matrices

$$\mathbf{M}_1 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{bmatrix} \quad \text{and} \quad \mathbf{M}_2 = \begin{bmatrix} \frac{1}{2} & 0 & \frac{\sqrt{3}}{2} \\ 0 & 1 & 0 \\ -\frac{\sqrt{3}}{2} & 0 & \frac{1}{2} \end{bmatrix}$$

respectively.

(a) Give a full geometrical description of the transformations:

(i) T_1 ;

(2)

(ii) T_2 .

(3)

(b) Find the matrix which represents the transformation T_1 followed by T_2 .

(2)

(c) The linear transformation T_3 is represented by the matrix

$$\mathbf{M}_3 = \begin{bmatrix} k & 2 & -1 \\ 1 & 1 & 1 \\ 3 & 4 & 1 \end{bmatrix}$$

where k is a constant.

For one particular value of k , T_3 has a line L of invariant points.

(i) Find k .

(ii) Find the Cartesian equations of L in the form $\frac{x}{p} = \frac{y}{q} = \frac{z}{r}$.

(7)

(Total 14 marks)

Q3.

Describe the single transformation represented by each of the matrices:

(a) $\begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix};$

(2)

(b) $\begin{bmatrix} 0.6 & 0 & -0.8 \\ 0 & 1 & 0 \\ 0.8 & 0 & 0.6 \end{bmatrix}.$

(3)

(Total 5 marks)

Q4.

(a) The transformation U of three-dimensional space is represented by the matrix

$$\begin{bmatrix} 1 & 4 & -3 \\ 2 & -1 & 0 \\ 1 & 1 & -1 \end{bmatrix}$$

(i) Write down a vector equation for the line L with cartesian equation

$$\frac{x-1}{2} = \frac{y-2}{3} = \frac{z-3}{6}$$

(2)

(ii) Find a vector equation for the image of L under U , and deduce that it is a line through the origin.

(4)

(b) The plane transformation V is represented by the matrix $\begin{bmatrix} 1 & 4 \\ 2 & -1 \end{bmatrix}.$

L_1 is the line with equation $y = \frac{1}{2}x + k$ and L_2 is the image of L_1 under V .

(i) Find, in the form $y = mx + c$, the cartesian equation for L_2

(4)

(ii) Deduce that L_2 is parallel to L_1 and find, in terms of k , the distance between these two lines.

(3)

(Total 13 marks)

Q5.

(a) Write down the 3×3 matrices which represent the transformations A and B , where:

(i) A is a reflection in the plane $y = x$; (2)

(ii) B is a rotation about the z -axis through the angle θ , where $\theta = \frac{\pi}{2}$. (1)

(b) (i) Find the matrix **R** which represents the composite transformation
'A followed by B' (3)

(ii) Describe the single transformation represented by **R**. (2)

(Total 8 marks)

Q6.

Give a full geometrical description of the transformation represented by each of the following matrices:

(a) $\begin{bmatrix} 0.8 & 0 & -0.6 \\ 0 & 1 & 0 \\ 0.6 & 0 & 0.8 \end{bmatrix}$; (3)

(b) $\begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$. (2)

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(Total 5 marks)

Q7.

A transformation **T** of three-dimensional space is given by the matrix $\mathbf{W} = \begin{bmatrix} 3 & -1 & 1 \\ 2 & 0 & 2 \\ -1 & 1 & 1 \end{bmatrix}$.

(a) (i) Evaluate $\det \mathbf{W}$, and describe the geometrical significance of the answer in relation to **T**. (2)

(ii) Determine the eigenvalues of **W**. (6)

(b) The plane **H** has equation $\mathbf{r} \cdot \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix} = 0$.

(i) Write down a cartesian equation for **H**.

(1)

- (ii) The point P has coordinates (a, b, c) . Show that, whatever the values of a, b and c , the image of P under T lies in H .

(4)

(Total 13 marks)

Q8.

The matrices $\mathbf{M}_A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & 1 & 0 \end{bmatrix}$ and $\mathbf{M}_B = \begin{bmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ represent the transformations A and B respectively.

- (a) Give a full geometrical description of each of A and B .

(5)

- (b) Transformation C is obtained by carrying out A followed by B .

- (i) Find $\mathbf{M}_C$, the matrix of C .

(2)

- (ii) Hence give a full geometrical description of the single transformation C .

(2)

(Total 9 marks)

Q9.

The matrices $\mathbf{A}$ and $\mathbf{B}$ are given by

$$\mathbf{A} = \begin{bmatrix} 1 & 1 \\ -1 & 1 \\ 1 & 1 \end{bmatrix} \quad \text{and} \quad \mathbf{B} = \begin{bmatrix} 1 & 0 & 2 \\ 2 & 2 & t \end{bmatrix}$$

- (a) Find, in terms of t , the matrices:

- (i) $\mathbf{AB}$;

(3)

- (ii) $\mathbf{BA}$.

(2)

- (b) Explain why $\mathbf{AB}$ is singular for all values of t .

(1)

- (c) In the case when $t = -2$, show that the transformation with matrix $\mathbf{BA}$ is the combination of an enlargement, E , and a second transformation, F . Find the scale factor of E and give a full geometrical description of F .

(6)

(Total 12 marks)

Q10.

Describe the geometrical transformation defined by the matrix

$$\begin{bmatrix} 0.6 & 0 & 0.8 \\ 0 & 1 & 0 \\ -0.8 & 0 & 0.6 \end{bmatrix}$$

(Total 3 marks)

Q11.

A transformation is represented by the matrix $\mathbf{A} = \begin{bmatrix} 0.28 & -0.96 & 0 \\ 0.96 & 0.28 & 0 \\ 0 & 0 & 1 \end{bmatrix}$.

- (a) Evaluate $\det \mathbf{A}$. (1)
- (b) State the invariant line of the transformation. (1)
- (c) Give a full geometrical description of this transformation. (3)

(Total 5 marks)

Q12.

The matrices $\mathbf{P}$ and $\mathbf{Q}$ are given by

$$\mathbf{P} = \begin{bmatrix} 2 & 1 & 1 \\ 1 & t & -2 \\ 3 & 2 & 1 \end{bmatrix} \quad \text{and} \quad \mathbf{Q} = \begin{bmatrix} 1 & 1 & 1 \\ -7 & -1 & 5 \\ 11 & -1 & -7 \end{bmatrix}$$

where t is a real constant.

- (a) Find the value of t for which $\mathbf{P}$ is singular. (2)
- (b) (i) Determine the matrix $\mathbf{R} = \mathbf{PQ}$, giving its elements in terms of t where appropriate. (3)
- (ii) Find the value of t for which $\mathbf{R} = k\mathbf{1}$, for some integer k . (2)
- (iii) Hence find the matrix $\mathbf{Q}^{-1}$. (1)
- (c) In the case when $t = -3$, describe the geometrical transformation with matrix $\mathbf{R}$. (2)

(Total 10 marks)

Q13.

- (a) Describe the transformation given by each of the matrices

$$\mathbf{A} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix} \quad \text{and} \quad \mathbf{B} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 1 & 1 \end{bmatrix}$$

(4)

- (b) (i) Determine the matrix $\mathbf{AB}$.

(2)

- (ii) Given that $\mathbf{AB}$ represents a rotation, find the axis of rotation and the magnitude of the angle of rotation.

(2)

(Total 8 marks)



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